CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

# Antifolate / nucleoside: 7 molecules stopped early, all small molecules

**20 stopped trials, but 7 molecules — all small molecules: Gemcitabine, Pemetrexed, Fluorouracil, Methotrexate and 3 more.** Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

**3.8%**
20 of 527 closed trials (95% CI 2.5%–5.8%)

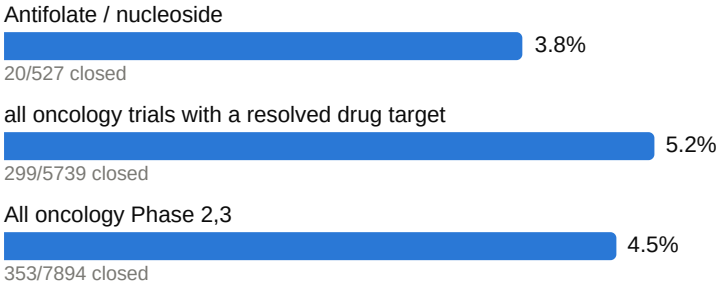
**12 / 7 / 1**
efficacy / safety / both (adds to 20)

**31.7%**
of 1665 trials have closed · 1138 still open or unresolved

**1.1%**
stopped for a biological reason within 3 years of starting (95% CI 0.6%–2.0%; 691 still at risk; 1.5% area-wide)

**7 of 11** **Completed and missed the primary endpoint.** Of 352 completed trials in this segment, 11 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 7 missed and 3 met, 1 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 182 of the 324 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

## How this compares




**What is in this class:** 549 distinct experimental drugs across 1665 trials (small molecule 1665, antibody 745, unknown 334, protein 130). Most tested: GEMCITABINE (496), CAPECITABINE (438), OXALIPLATIN (393), FLUOROURACIL (388), PACLITAXEL (294). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 20 stops came from 20 sponsor-asset programmes across 20 sponsors; the largest (Universität Duisburg-Essen) contributed 1; removing that programme's trials from both sides leaves 3.6%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 44 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 12.1%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI |
|---|---|---|---|---|
| 2015–2017 | 12 | 239 | 5.0% | 2.9%–8.6% |
| 2018–2020 | 4 | 171 | 2.3% | 0.9%–5.9% |
| 2021–2024 | 4 | 117 | 3.4% | 1.3%–8.5% |

## Where the stops sit

Efficacy / futility: 12, Safety: 7, Efficacy and safety: 1.
Sponsors with most stops: University of Nebraska (1), BerGenBio ASA (1), Tufts Medical Center (1), Roswell Park Cancer Institute (1), Ukrainian Society of Clinical Oncology (1), AHS Cancer Control Alberta (1).

## The stopped trials

| Trial | Ph | Sponsor | Experimental drugs | Stopped | Type | Registry stop reason |
|---|---|---|---|---|---|---|
| NCT04141995 | 2 | University of Nebraska | DIGOXIN, FLUOROURACIL, CALCIUM, IRINOTECAN | 2025-05 | Efficacy | Due to low accrual and futility |
| NCT05469178 | 2 | BerGenBio ASA | BEMCENTINIB, PEMBROLIZUMAB, PEMETREXED, CARBOPLATIN | 2025-04 | Efficacy | Lack of efficacy |
| NCT04856176 | 2 | Tufts Medical Center | Granulocyte-Macrophage Colony-Sti…, PEMBROLIZUMAB, PEMETREXED, PACLITAXEL | 2024-12 | Efficacy | Futility |
| NCT03333486 | 2 | Roswell Park Cancer Institute | CYCLOPHOSPHAMIDE ANHYDROUS, FLUDARABINE PHOSPHATE | 2023-08 | Efficacy | Futility analysis determined closure |
| NCT04937738 | 2 | Ukrainian Society of Clinical Oncology | DOCETAXEL ANHYDROUS, OXALIPLATIN, LEUCOVORIN, FLUOROURACIL | 2023-02 | Efficacy | High rate of disease progression in XELOX arm |
| NCT02295722 | 2 | AHS Cancer Control Alberta | GEMCITABINE, MELPHALAN | 2023-02 | Efficacy | It did not show a significant benefit to justify completing the full target accrual. |

The 6 most recent of 20 stopped trials. Every molecule behind all 20 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

**Method**
Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 31.7% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**The whole cohort, trial by trial.** The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.